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Idea Submission Template / Code. Create. Innovate.

Team Name	Team Raksha
Problem Statement No.	2
Problem Statement Title	Improving Safety at Large Public Events
Team Leader Name	Shounak Joshi

THE PROBLEM

Guards watch screens. Danger builds unseen.

A control room watches dozens of flat camera feeds. A guard spots trouble only after the crowd stops moving. By then, crushing pressure is already passing body to body. Stillness is not the warning. It is the accident.

Hathras 2024

121 dead. 80k permit, 250k crowd. Exits blocked.

Mumbai 2017

23 dead on a station footbridge. Rain, rush hour, no metering.

7 per m2

A crowd turns fluid. Shockwaves throw people metres. (Fruin)

Why stillness lies

A stopped crowd is not a warning. It is the accident. Pressure passes body to body long before feet stop. That is why cameras must read force, not faces.

Kumbh, 29 January 2025



Months before

2,751 cameras and AI alerts go live.



01:00

Barricades jumped at the Sangam nose. Crores press forward.



After

30 dead officially. The alarm watched. Nobody acted.



Kumbh mela ground at night, Jan 2025. Photo: Wikimedia Commons.

The system saw it coming. No gate opened.

Danger is invisible pressure, not visible stillness.

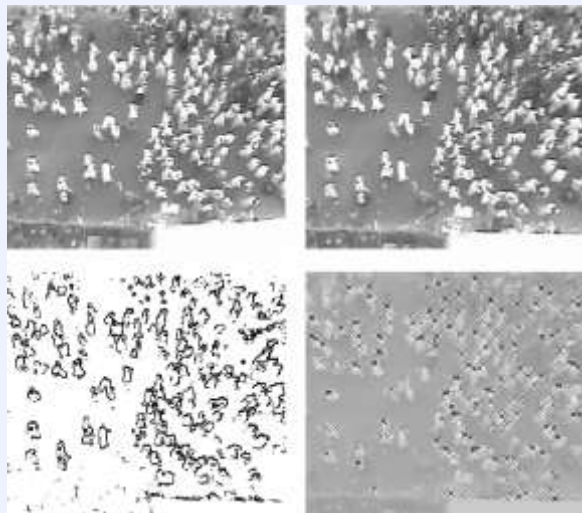
Measure pressure, warn early, open gates.

HOW IT WORKS



In plain words: map the ground right, count the crowd as heat, read its currents, warn on pressure.

What the machine sees



Head-density surface from video (Johansson 2008).



Flow field on our station clip (DIS optical flow). Movers glow, stillness stays black.

Heatmaps beat boxes

At crush density a detector scores F1 0.07 against 0.71 point-based. Counting error fell from 68 to 49 MAE. (NWPU-Crowd, ShanghaiTech)

Pressure, not stillness

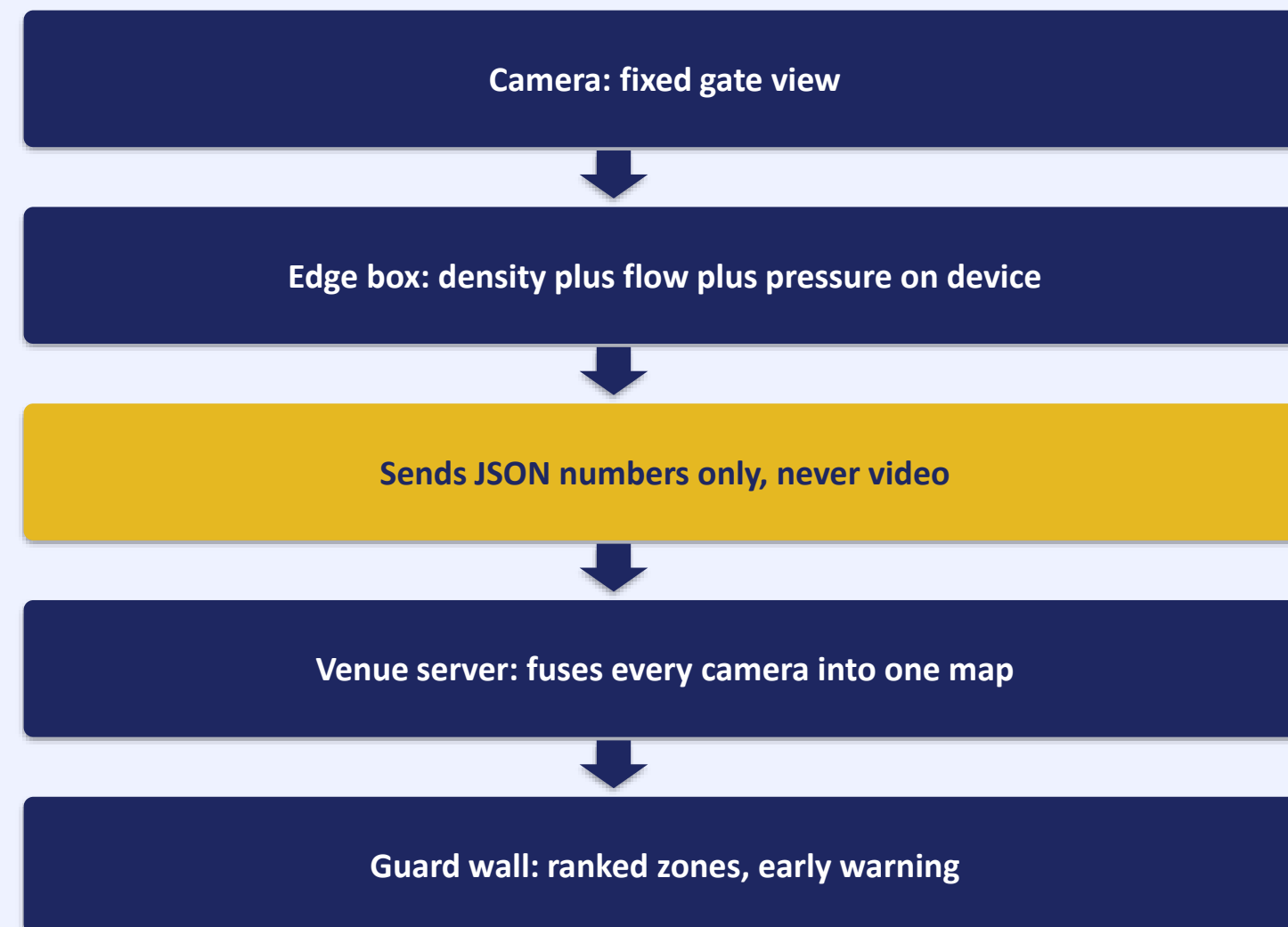
Pressure is density times uneven movement. Density or speed alone cannot warn. (Johansson 2008)

Warns before stillness

Stop-and-go waves and rising pressure fire the alert while feet still move. Stillness is the accident, never the trigger.

FEASIBILITY AND VIABILITY

One camera, one edge box



Raw video never leaves the camera box.

Video stays put

Each edge box keeps a 10-minute rolling buffer and sends small JSON numbers, never raw video. Only a confirmed Red alert exports its clip, signed, to the evidence pack.

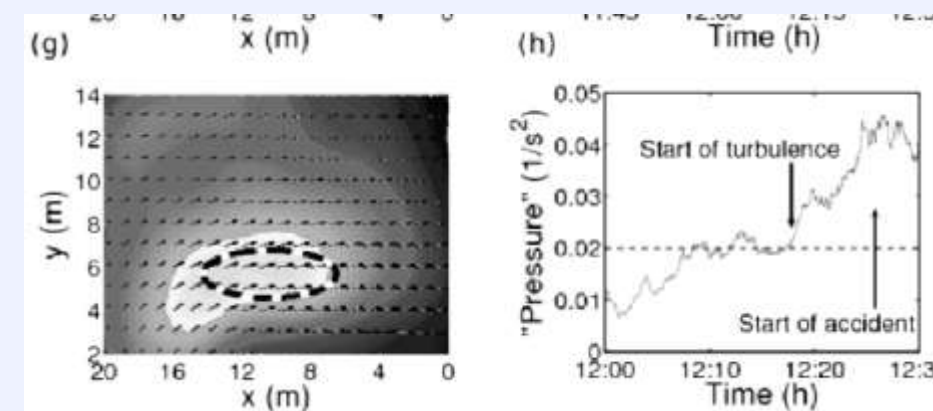
Calibrated where it counts

Fixed gates are measured once. Kumbh ran 3-month camera surveys for its crowd alarm. PTZ or moved cameras fall back to relative levels until re-measured.

Private by design

Heatmaps cannot show a face. Saved video is blurred, wiped after 7 days, zero identity in core.

Pressure warned first



Pressure crossed the warning line before turbulence began. The accident came after. Hajj 2006 data (Johansson et al.).

IMPACT AND BENEFITS

When pressure rises

Yellow watch

Guards watch the zone closely. Overflow gates get ready.

Red alert

Guards get steps on their handsets; a human confirms before any PA or door release.

Handover

A sealed evidence pack goes to the police.

Pilot targets: warn before critical, page guards in seconds, log every step.

Before

A wall of screens, one tired guard, and risks get missed.

After

Pressure rising at Gate B opens overflow at Gate A upstream. Security acts before the jam, not after.

Organizers

Spot a growing jam early, and tell guards which zone to clear. Built for stations, mela grounds, stadiums.

Police

Every alert, zone and action exports as one evidence pack in a click, with the signed 10-minute incident clip.

Public

Faces stay blurred in saved video.

RESEARCH AND REFERENCES

The science behind it

Count the overlap

[CSRNet, CVPR 2018](#)

[Bayesian Loss, ICCV 2019](#)

[DM-Count, NeurIPS 2020](#)

[P2PNet, ICCV 2021](#)

[STEERER / APGCC, 2023-24](#)

Tests: ShanghaiTech, QNRF, NWPU

Build from: [DM-Count code \(MIT\)](#) [CSRNet model](#) [homography calibrator](#) [DeepStream custom-model docs](#)

Ground and currents

[Chan perspective maps, CVPR 2008](#)

[Zhang ground-plane fusion, CVPR 2019](#)

[PACNN perspective, CVPR 2019](#)

[Mehran social force, CVPR 2009](#)

[Mehran streaklines, ECCV 2010](#)

[DIS optical flow \(OpenCV\)](#)

Pressure and risk

[Helbing social force, 1995](#)

[Helbing crowd disasters, 2007](#)

[Johansson pressure, 2008](#)

[Hughes continuum, 2002](#)

[Nakayama divergence, BMVC 2025](#)

[Harding early warning, PLoS ONE 2011](#)

Disasters, limits, footage

Itaewon and Love Parade

159 dead, Seoul 2022, ignored calls. 21 dead, Duisburg 2010. Turbulence, not panic.

Kumbh and Hathras

30 dead, Jan 2025, with AI live. 121 dead, Jul 2024, 80k permit vs 250k crowd.

Danger thresholds

Fluid at 7 per m2. Serious above 4. Flow peaks at 2 to 3. (Fruin, Still)

Test footage

Public clips: station, crossing, temple.
[Wikimedia Commons](#)